

Lab Experiment 12

Tic Tac Toe Game

Aim

To write a Python program to implement the Tic Tac Toe game for two players.

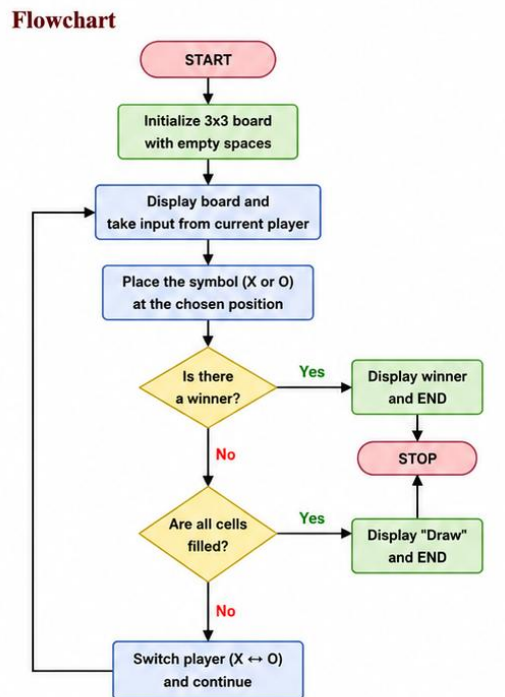
Objective

The objective of this experiment is to develop a simple Tic Tac Toe game using Python. It aims to allow two players to play alternately and identify the winner or draw.

Algorithm

1. Create a 3×3 board with empty spaces.
2. Display the board and take input from the current player.
3. Place **X** or **O** in the selected position.
4. Check whether the current player has won.
5. If a player wins, display the winner and end the game.
6. If all cells are filled, declare the game as a draw.
7. Otherwise, switch the player and continue the game.

Flowchart



Code

```
def print_board(board):
```

```
    for row in board:
```

```
        print(" | ".join(row))
```

```
    print("-" * 9)
```

```
def check_winner(board, player):
```

```
    # Check rows and columns
```

```
    for i in range(3):
```

```
        if all(board[i][j] == player for j in range(3)):
```

```
            return True
```

```
        if all(board[j][i] == player for j in range(3)):
```

```
            return True
```

```
# Check diagonals
```

```
if all(board[i][i] == player for i in range(3)):
```

```
    return True
```

```
if all(board[i][2 - i] == player for i in range(3)):
```

```
    return True
```

```
return False
```

```
def is_draw(board):
```

```
    return all(cell != ' ' for row in board for cell in row)
```

```
def tic_tac_toe():
```

```
    board = [[' ' for _ in range(3)] for _ in range(3)]
```

```
    players = ['X', 'O']
```

```
    turn = 0
```

```
    while True:
```

```
        print_board(board)
```

```
        player = players[turn]
```

```
        row, col = map(
```

```
            int,
```

```
            input(f'Player {player}, enter row and column (0-2): ').split()
```

```
        )
```

```
        if board[row][col] != ' ':
```

```

        print("Cell already occupied! Try again.")
        continue

    board[row][col] = player

    if check_winner(board, player):
        print_board(board)
        print(f"Player {player} wins!")
        break

    if is_draw(board):
        print_board(board)
        print("It's a Draw!")
        break

    turn = 1 - turn

tic_tac_toe()

```

Output

```

Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/ABBIRAMY/Documents/AI-lab 12.py =====
  | |
--
  | |
--
  | |
--
Player X, enter row and column (0-2): |

```

Result

The Python program successfully implements the Tic Tac Toe game and correctly identifies the winner or draw.

Conclusion

The Tic Tac Toe program demonstrates the use of loops, functions, conditional statements, and lists to develop a simple two-player game in Python.